

DEVILS LAKE, ND, USA

WMO: 727573

Lat: 48.117N

Lon: 98.900W

Elev: 439

StdP: 96.16

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94928

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.1	(c) -27.1	(d) -33.0	(e) 0.2	(f) -26.7	(g) -32.1	(h) 0.2	(i) -25.6	(j) 14.8	(k) -10.9	(l) 13.8	(m) -11.3	(n) 3.6	(o) 260	(p) 0.712

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	30.1	21.1	28.0	19.7	27.2	19.0	22.7	27.9	21.4	26.9	20.2	25.4	5.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.1	16.6	26.4	19.1	14.6	24.9	17.9	13.6	23.5	69.2	27.9	64.1	26.9	60.0	25.6	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.9	(b) 11.6	(c) 10.4	(e) -32.2	(f) 33.7	(g) 3.2	(h) 1.8	(i) -34.5	(j) 34.9	(k) -36.4	(l) 36.0	(m) -38.3	(n) 37.0	(o) -40.6	(p) 38.3		
(a) 12.9	(b) 11.6	(c) 10.4	(e) -31.4	(f) 24.8	(g) 2.4	(h) 1.5	(i) -33.2	(j) 25.9	(k) -34.6	(l) 26.8	(m) -35.9	(n) 27.7	(o) -37.6	(p) 28.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.4	-13.6	-11.9	-4.4	4.2	11.8	17.6	20.4	19.5	14.5	6.2	-2.4	-10.6
	DBStd	13.43	8.09	7.58	7.35	5.93	4.83	3.72	3.14	3.38	4.62	5.13	6.26	7.35
	HDD10.0	3181	733	613	447	187	37	1	0	0	12	138	374	638
	HDD18.3	5299	991	847	705	424	209	57	16	27	131	375	623	896
	CDD10.0	1122	0	0	1	13	93	228	324	293	148	22	1	0
	CDD18.3	198	0	0	0	0	6	33	81	62	16	0	0	0
	CDH23.3	1576	0	0	0	3	68	258	612	486	142	7	0	0
	CDH26.7	387	0	0	0	0	13	53	153	134	32	1	0	0
Wind	WSAvg	5.0	5.1	5.1	5.2	5.3	5.5	4.9	4.4	4.4	4.8	5.4	5.3	5.1
Precipitation	PrecAvg	442	14	12	22	27	55	84	67	53	48	28	17	15
	PrecMax	706	32	36	82	89	159	168	206	106	133	125	99	31
	PrecMin	255	0	0	4	0	8	22	8	6	4	0	1	2
	PrecStd	96	9	9	17	19	39	40	42	29	34	27	19	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.5	5.2	16.1	22.6	28.3	30.9	32.5	32.7	30.0	23.9	15.7	5.2
		MCWB	1.3	2.5	9.7	11.9	16.8	20.5	22.7	22.5	20.3	15.0	9.4	2.0
	2%	DB	2.2	2.7	11.1	18.9	25.1	27.9	30.0	29.9	27.2	19.1	11.2	2.8
		MCWB	0.6	1.1	7.1	10.3	14.8	19.0	22.0	20.8	18.2	12.6	6.6	0.6
	5%	DB	0.2	1.3	7.5	16.3	22.6	26.3	28.0	27.8	24.7	17.0	8.0	1.2
		MCWB	-0.8	-0.1	4.4	8.7	13.5	17.8	20.4	19.7	16.7	10.7	4.5	-0.3
	10%	DB	-2.3	-1.0	4.1	13.0	20.3	24.1	27.1	26.2	22.4	13.8	6.0	-1.1
		MCWB	-3.2	-1.9	2.1	7.4	12.4	16.9	19.7	18.7	15.8	9.2	3.0	-2.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.6	3.0	10.7	12.9	19.0	23.0	24.7	24.1	21.4	16.1	9.9	2.6
		MCDB	3.2	4.6	13.8	20.8	25.3	28.5	29.8	30.1	27.3	21.8	14.9	5.1
	2%	WB	0.6	1.2	7.2	10.9	16.7	20.7	23.0	22.1	19.6	13.4	6.9	0.9
		MCDB	2.0	2.8	10.8	17.6	22.4	26.3	27.8	27.7	25.0	18.6	10.5	2.6
	5%	WB	-0.9	0.0	4.5	9.6	14.8	19.4	21.7	20.7	18.0	11.3	4.9	-0.3
		MCDB	0.1	1.3	7.3	15.6	20.2	24.4	26.9	26.6	23.0	16.0	8.0	1.1
	10%	WB	-3.1	-1.9	2.4	7.9	13.4	18.0	20.5	19.7	16.5	9.6	3.0	-2.0
		MCDB	-2.2	-1.0	4.5	12.8	18.9	22.9	25.8	24.8	21.2	13.9	5.7	-1.1
Mean Daily Temperature Range	5% DB	MDBR	8.9	9.2	8.8	10.3	10.9	9.7	10.0	10.6	10.6	9.0	8.0	7.9
		MCDBR	10.0	9.2	12.0	15.0	14.3	12.2	11.6	12.6	13.9	14.0	11.6	9.5
		MCWBR	9.1	7.9	8.4	8.1	7.0	5.9	5.9	6.2	7.2	8.2	7.9	8.0
	5% WB	MCDBR	9.9	8.8	11.9	13.8	11.9	10.6	10.6	11.5	12.2	12.9	11.3	9.4
		MCWBR	9.1	7.5	8.6	8.0	6.6	6.2	6.2	6.4	7.2	8.1	7.9	8.0
Clear-Sky Solar Irradiance	taub	0.286	0.292	0.294	0.361	0.373	0.377	0.381	0.384	0.351	0.297	0.268	0.251	
	taud	2.252	2.278	2.443	2.310	2.341	2.378	2.378	2.355	2.425	2.552	2.498	2.430	
	Ebn at Noon	775	865	932	889	889	885	875	856	855	863	821	800	
	Edh at Noon	76	93	93	119	120	117	115	113	95	71	59	56	
All-Sky Solar Radiation	RadAvg	1.61	2.76	3.94	4.95	5.64	6.12	6.46	5.36	3.97	2.48	1.48	1.13	
	RadStd	0.13	0.27	0.35	0.39	0.36	0.42	0.33	0.36	0.26	0.31	0.20	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page